

Uniqueness Without Monotonicity: Scalar Dynamics for Spectral Transforms

Candidate full answer to Appendix A, Problem 1 of arXiv:2508.04890

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Status. Candidate full answer, likely valid, subject to human review. The source’s phrase “order of spectral projections” is not defined; the result treats both the standard spectral/Loewner order and the source’s formally stated projection-stability axiom.

1 Source problem

The target is Appendix A, Problem 1 (PDF p. 11) in Alpay–Alakkad–Alpay, *Transfinite Operator Fixed Points on Hilbert Spaces: An Alpay Algebra Approach*, arXiv:2508.04890v1 [1]. It asks whether the monotonicity assumption is essential, requests a non-order-preserving transformation with a unique iterated limit, and alternatively asks for weaker hypotheses that guarantee uniqueness.

algebras and contribute to the categorical and logical aspects of Alpay Algebra, enriching both domains through this interdisciplinary connection.

A Open Problems and Further Directions

We list several open problems that arise from our study, ordered roughly from more accessible to increasingly speculative. Each problem extends the ideas in this paper, layering on additional complexity or generality, and in doing so, each becomes correspondingly harder – forming a chain of challenges that grows denser and more formidable as one progresses.

Problem 1: Uniqueness without Monotonicity. In Theorem 3.1, we assumed a form of monotonicity for Φ to ensure uniqueness of the fixed point A_∞ . Is this assumption essential? Open sub-problem: Construct (or rule out) an example of a transformation Φ that does not preserve the order of spectral projections, yet still yields a unique $\Phi^\infty(A)$. Conversely, find conditions weaker than monotonicity that guarantee uniqueness. This problem asks for a refinement of our convergence theory: without monotonicity, one might get multiple fixed points or cycling behavior, complicating the “limit” concept. Tackling this would clarify the necessity of certain assumptions and possibly extend the applicability of the fixed-point theorem.

Problem 2: Transfinite Extension of Banach’s Fixed-Point Theorem. Banach’s contraction principle guarantees a unique fixed point for a contraction mapping on a complete metric space in a single step limit. In our context, Φ might not be a contraction in operator norm (it could even increase norms), yet Φ^n can converge in a transfinite sense. Question: Can

Figure 1: Appendix A, Problem 1 in the source paper (PDF p. 11).

The source’s formal definition uses a stability condition that, in particular, requires $\Phi(P) = P$ for every orthogonal projection P . Its appendix instead refers to preservation of an undefined “order of spectral projections.” We resolve both natural readings below.

2 Definitions and result

Fix a nonzero complex Hilbert space H , a compact set $K \subset \mathbb{R}$, and

$$\mathcal{S}_K(H) = \{A \in B(H) : A = A^*, \sigma(A) \subseteq K\}.$$

For $f \in C(K, \mathbb{C})$, let $\Phi_f(A) = f(A)$, using the continuous functional calculus. Write $f^{\circ n}$ for scalar composition. For reference, the spectral order may be defined by

$$A \preceq_s B \iff E_A((t, \infty)) \leq E_B((t, \infty)) \quad (t \in \mathbb{R}).$$

It agrees with the ordinary order on scalar multiples of the identity.

Proof intuition

A spectral transform does not need an order structure to converge. Every operator orbit is obtained by running the same one-dimensional dynamical system $t \mapsto f(t)$ at each spectral value. If the scalar orbits converge, the spectral theorem integrates those pointwise limits; bounded convergence turns them into strong operator convergence. Thus uniqueness is controlled by scalar attraction, not by monotonicity. Squaring on $[-1, 1]$ is already nonmonotone but converges pointwise to the indicator of $\{-1, 1\}$. Damping the square by $1/2$ makes zero a uniform global attractor and even destroys the source's projection stability.

Theorem 1 (Scalar-orbit criterion). *Let $K \subset \mathbb{R}$ be compact, $f \in C(K, \mathbb{C})$, and suppose that*

$$f^{\circ n}(t) \longrightarrow g(t) \quad (t \in K).$$

The pointwise limit g is Borel, so $g(A)$ is defined by the bounded Borel functional calculus. Then, for every $A \in \mathcal{S}_K(H)$,

$$\Phi_f^n(A) = f^{\circ n}(A) \xrightarrow[n \rightarrow \infty]{\text{SOT}} g(A), \quad \Phi_f(g(A)) = g(A).$$

Consequently the orbit limit is unique and is reached at the first limit stage ω ; no order-monotonicity assumption is needed. If the scalar convergence is uniform on K , the operator convergence is in norm.

If all scalar orbits have one common limit $\xi \in K$, then $\Phi_f^n(A) \rightarrow \xi I$ strongly for every A , and ξI is the unique fixed point of Φ_f in $\mathcal{S}_K(H)$. Conversely, convergence to the same scalar operator ξI for every A forces $f^{\circ n}(t) \rightarrow \xi$ for every $t \in K$.

Proof. Functoriality of the continuous functional calculus gives $\Phi_f^n(A) = f^{\circ n}(A)$. Let E_A be the spectral measure of A and $\mu_x(\Delta) = \langle E_A(\Delta)x, x \rangle$. Since every iterate and the pointwise limit take values in the bounded set K , dominated convergence gives, for every $x \in H$,

$$\|(f^{\circ n}(A) - g(A))x\|^2 = \int_K |f^{\circ n}(t) - g(t)|^2 d\mu_x(t) \longrightarrow 0.$$

This is strong convergence. If the convergence on K is uniform, then

$$\|f^{\circ n}(A) - g(A)\| \leq \sup_{t \in K} |f^{\circ n}(t) - g(t)| \longrightarrow 0.$$

For each t , continuity of f yields

$$f(g(t)) = \lim_n f(f^{\circ n}(t)) = \lim_n f^{\circ(n+1)}(t) = g(t).$$

Hence $f \circ g = g$, and functional calculus gives $\Phi_f(g(A)) = g(A)$.

Now suppose $g \equiv \xi$. The preceding argument gives the common limit ξI . If $B = f(B)$, then the continuous function $h(t) = t - f(t)$ satisfies $h(B) = 0$, so h vanishes on $\sigma(B)$. Every $t \in \sigma(B)$ is therefore a scalar fixed point. Its scalar orbit is constant and also must converge to ξ , whence $t = \xi$. Thus $\sigma(B) \subseteq \{\xi\}$ and $B = \xi I$. Conversely, apply the claimed common operator convergence to the scalar operator $A = tI$. $\square$

3 Two explicit answers to Problem 1

Proposition 1 (The source’s canonical squaring map). *On $\mathcal{S}_{[-1,1]}(H)$, define $\Phi(A) = A^2$. This transformation is neither Loewner-order nor spectral-order preserving, yet every orbit has the unique strong limit*

$$\Phi^n(A) = A^{2^n} \xrightarrow[n \rightarrow \infty]{\text{SOT}} E_A(\{-1, 1\}).$$

The limit is an orthogonal projection and hence is fixed by Φ .

Proof. For $f(t) = t^2$ on $[-1, 1]$,

$$f^{\circ n}(t) = t^{2^n} \longrightarrow g(t) = \mathbf{1}_{\{-1, 1\}}(t).$$

Theorem 1 gives the stated strong limit. It is fixed because its square equals itself. Nonmonotonicity already occurs on scalar operators: $-I \leq 0$ and $-I \preceq_s 0$, but $\Phi(-I) = I$ is neither $\leq 0 = \Phi(0)$ nor $\preceq_s 0$. $\square$

This also corrects the source’s motivating claim: for self-adjoint contractions the limit is generally $E_A(\{-1, 1\})$, not only the projection onto $\ker(I - A)$. The one-dimensional case $A = -I$ makes the difference immediate.

Proposition 2 (Failure of projection stability with a unique global fixed point). *On $\mathcal{S}_{[-1,1]}(H)$, define $\Psi(A) = \frac{1}{2}A^2$. Then Ψ is neither Loewner-order nor spectral-order preserving and violates the source’s formal stability axiom: $\Psi(P) = P/2 \neq P$ for every nonzero orthogonal projection P . Nevertheless*

$$\Psi^n(A) = 2^{1-2^n} A^{2^n}, \quad \|\Psi^n(A)\| \leq 2^{1-2^n} \longrightarrow 0.$$

Thus 0 is the norm limit of every orbit and is the unique fixed point in the self-adjoint unit ball.

Proof. The scalar identity follows by induction from $h(t) = t^2/2$, and the norm estimate uses $\|A\| \leq 1$. If $B = \Psi(B)$ and $r = \|B\|$, then $r \leq r^2/2$. For $0 < r \leq 1$ this is impossible, so $r = 0$ and $B = 0$. The projection calculation is explicit. Finally, $-I \leq 0$ (also in spectral order), while $\Psi(-I) = I/2$ lies above $\Psi(0) = 0$, proving both order failures. $\square$

Together Propositions 1 and 2 answer both parts of Problem 1: they provide explicit non-order-preserving transforms with unique iterated limits, while Theorem 1 supplies a weaker, purely dynamical condition guaranteeing uniqueness.

4 Verification and limitations

The proof is dimension-free and includes continuous spectrum; it does not assume diagonalizability. The included script `code/check_examples.py` checks the finite-spectrum limits, the exact damped-iterate formula through seven stages, and the scalar order failure. It is a sanity check, not evidence replacing the proof.

There are three scope qualifications.

1. The source never defines “order of spectral projections.” We use the standard spectral order and separately violate its formally stated projection-stability condition.
2. “Unique $\Phi^\infty(A)$ ” is read as a uniquely determined orbit limit for each initial A . Proposition 2 also gives the stronger property of a unique global fixed point.
3. The result is in the fixed-H, bounded self-adjoint spectral-transform setting. This is a special case of the source’s framework and suffices for the requested construction; it does not repair the paper’s broader transfinite claims.

Novelty search. On 17 August 2026, bounded searches covered the run’s registry, solution, attempt, and proof-gap indexes, plus arXiv searches by exact title, arXiv id, Problem 1 label, the quoted order phrase, and close uniqueness/spectral keywords. Only the source paper was located; no later explicit answer or correction was found. Because the proof is an elementary application of the spectral theorem, novelty confidence is moderate rather than high.

Human-review recommendation. Check the intended meanings of the source’s undefined order phrase and “unique limit,” then verify the common-attractor fixed-point argument in Theorem 1. The mathematical computations themselves are short and should be reproducible directly from the spectral theorem.

References

- [1] F. Alpay, H. Alakkad, and T. Alpay, *Transfinite Operator Fixed Points on Hilbert Spaces: An Alpay Algebra Approach*, arXiv:2508.04890v1 (2025), Appendix A, Problem 1.